Step-by-step deployment procedure

NetApp Solutions

NetApp
June 16, 2021

This PDF was generated from https://docs.netapp.com/us-en/netapp-solutions/ent-apps-db/awx_automation.html on July 11, 2021. Always check docs.netapp.com for the latest.

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Step-by-step deployment procedure

AWX/Tower deployment Oracle 19c Database

1. Create the inventory, group, hosts, and credentials for your environment

This section describes the setup of inventory, groups, hosts, and access credentials in AWX/Ansible Tower that prepare the environment for consuming NetApp automated solutions.

1. Configure the inventory.
   a. Navigate to Resources → Inventories → Add, and click Add Inventory.
   b. Provide the name and organization details, and click Save.
   c. On the Inventories page, click the inventory created.
   d. If there are any inventory variables, paste them in the variables field.
   e. Navigate to the Groups sub-menu and click Add.
   f. Provide the name of the group for ONTAP, paste the group variables (if any) and click Save.
   g. Repeat the process for another group for Oracle.
   h. Select the ONTAP group created, go to the Hosts sub-menu and click Add New Host.
   i. Provide the IP address of the ONTAP cluster management IP, paste the host variables (if any), and click Save.
   j. This process must be repeated for the Oracle group and Oracle host(s) management IP/hostname.
2. Create credential types. For solutions involving ONTAP, you must configure the credential type to match username and password entries.
   a. Navigate to Administration → Credential Types, and click Add.
   b. Provide the name and description.
   c. Paste the following content in Input Configuration:

```
fields:
  - id: username
    type: string
    label: Username
  - id: password
    type: string
    label: Password
    secret: true
  - id: vsadmin_password
    type: string
    label: vsadmin_password
    secret: true
```

   d. Paste the following content into Injector Configuration:

```
extra_vars:
  password: '{{ password }}'
  username: '{{ username }}'
  vsadmin_password: '{{ vsadmin_password }}'
```

3. Configure the credentials.
   a. Navigate to Resources → Credentials, and click Add.
   b. Enter the name and organization details for ONTAP.
   c. Select the custom Credential Type you created for ONTAP.
   d. Under Type Details, enter the username, password, and vsadmin_password.
   e. Click Back to Credential and click Add.
   f. Enter the name and organization details for Oracle.
   g. Select the Machine credential type.
   h. Under Type Details, enter the Username and Password for the Oracle hosts.
   i. Select the correct Privilege Escalation Method, and enter the username and password.

2. Create a project

1. Go to Resources → Projects, and click Add.
   a. Enter the name and organization details.
   b. Select Git in the Source Control Credential Type field.
   c. enter https://github.com/NetApp-Automation/na_oracle19c_deploy.git as the source control URL.
   d. Click Save.
   e. The project might need to sync occasionally when the source code changes.

3. Configure Oracle host_vars

The variables defined in this section are applied to each individual Oracle server and database.

1. Input your environment-specific parameters in the following embedded Oracle hosts variables or host_vars form.

> The items in blue must be changed to match your environment.

Unresolved directive in ent-apps-db/awx_automation.adoc - include::ent-apps-db/host_vars.adoc[]

   a. Fill in all variables in the blue fields.
   b. After completing variables input, click the Copy button on the form to copy all variables to be transferred to AWX or Tower.
   c. Navigate back to AWX or Tower and go to Resources → Hosts, and select and open the Oracle server configuration page.
   d. Under the Details tab, click edit and paste the copied variables from step 1 to the Variables field under the YAML tab.

e. Click Save.

f. Repeat this process for any additional Oracle servers in the system.

4. Configure global variables

Variables defined in this section apply to all Oracle hosts, databases, and the ONTAP cluster.

1. Input your environment-specific parameters in following embedded global variables or vars form.

   The items in blue must be changed to match your environment.

   Unresolved directive in ent-apps-db/awx_automation.adoc - include::ent-apps-db/vars.adoc[]

2. Fill in all variables in blue fields.
3. After completing variables input, click the Copy button on the form to copy all variables to be transferred to AWX or Tower into the following job template.

5. Configure and launch the job template.

1. Create the job template.
   a. Navigate to Resources → Templates → Add and click Add Job Template.
   b. Enter the name and description
   c. Select the Job type; Run configures the system based on a playbook, and Check performs a dry run of a playbook without actually configuring the system.
   d. Select the corresponding inventory, project, playbook, and credentials for the playbook.
   e. Select the all_playbook.yml as the default playbook to be executed.
   f. Paste global variables copied from step 4 into the Template Variables field under the YAML tab.
   g. Check the box Prompt on Launch in the Job Tags field.
   h. Click Save.
2. Launch the job template.
   a. Navigate to Resources → Templates.
   b. Click the desired template and then click Launch.
   c. When prompted on launch for Job Tags, type in requirements_config. You might need to click the Create Job Tag line below requirements_config to enter the job tag.

      requirements_config ensures that you have the correct libraries to run the other roles.

   d. Click Next and then Launch to start the job.
   e. Click View → Jobs to monitor the job output and progress.
   f. When prompted on launch for Job Tags, type in ontap_config. You might need to click the Create "Job Tag" line right below ontap_config to enter the job tag.
   g. Click Next and then Launch to start the job.
   h. Click View → Jobs to monitor the job output and progress

i. After the ontap_config role has completed, run the process again for linux_config.
j. Navigate to Resources → Templates.
k. Select the desired template and then click Launch.
l. When prompted on launch for the Job Tags type in linux_config, you might need to select the Create "job tag" line right below linux_config to enter the job tag.
m. Click Next and then Launch to start the job.
n. Select View → Jobs to monitor the job output and progress.
o. After the linux_config role has completed, run the process again for oracle_config.
p. Go to Resources → Templates.
q. Select the desired template and then click Launch.
r. When prompted on launch for Job Tags, type oracle_config. You might need to select the Create "Job Tag" line right below oracle_config to enter the job tag.
s. Click Next and then Launch to start the job.
t. Select View → Jobs to monitor the job output and progress.

6. Deploy additional database on same Oracle host

The Oracle portion of the playbook creates a single Oracle container database on an Oracle server per execution. To create additional container databases on the same server, complete the following steps.

1. Revise host_vars variables.
   a. Go back to step 2 - Configure Oracle host_vars.
   b. Change the Oracle SID to a different naming string.
   c. Change the listener port to different number.
   d. Change the EM Express port to a different number if you are installing EM Express.
   e. Copy and paste the revised host variables to the Oracle Host Variables field in the Host Configuration Detail tab.
2. Launch the deployment job template with only the oracle_config tag.

Unresolved directive in ent-apps-db/awx_automation.adoc - include::ent-apps-db/validation.adoc[]

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